Decomposition of leaf litter in headwater streams.

Effects of changes in the environment and contribution of microbial and shredder activity on litter decomposition.

Johan Lidman

Degree thesis in ecology 60 hp
Master's level
Report passed: 16 06 2015
Supervisor: Micael Jonsson

Decomposition of leaf litter in headwater streams.

Effects of changes in the environment and contribution of microbial and shredder activity on litter decomposition.

Johan Lidman

Abstract

Headwaters, which are the most common stream order in the landscape, are mostly dependent on energy produced in the terrestrial system, largely consisting of leaf litter from riparian vegetation. The aim of this study was to investigate the decomposition in headwaters of leaf litter from three native (alder, birch, spruce) and one non-native (lodgepole pine) species and how decomposition responds to changes in the environment. Further, microbial and shredder influences on leaf-litter decomposition and aquatic decomposer ability to adapt to non-native species was investigated. By using field-data from this study, calculations were made to assess if microbes and shredders are resource limited. Litterbags were placed in 20 headwater streams in northern Sweden that varied in water chemistry, stream physical characteristics and riparian vegetation. The results revealed that species litter decomposition of different plant species was affected differently by changes in environmental variables. Alder and birch decomposition were positively associated, whereas lodgepole pine deviated from the other species in decomposition and its relationship with important environmental variables, indicating that the ability of the boreal aquatic systems to decompose litter differs between introduced and native species. When including macroinvertebrates, shredder fragmentation generally increased decomposition, but was not significant for all sites. Resource availability for microbes and shredders was controlled by litter input, and no risk of resource limitations was evident during the study period. These findings highlight a complexity of the decomposition process that needs to be considered when predicting changes due to human activities.

Key words: Leaf-litter decomposition, headwaters, microbial decomposition, shredders, ecosystem adaptation

Contents

1 Introduction 1
1.1 Headwaters 1
1.2 Decomposition of litter 1
1.3 Litter quality and stoichiometry 2
1.4 Macroinvertebrate production 2
1.5 Human disturbance 2
1.6 Study aims 3
2 Material and methods 3
2.1 Study site 3
2.2 Water chemistry and stream characteristics 4
2.3 Litter decomposition 5
2.3.1 Study design 5
2.3.2 Mass loss 5
2.3.3 Decomposition rate 5
2.3.4 Nutrient content 6
2.4 Litter input 6
2.5 Estimation of litter standing stock 6
2.6 Macroinvertebrates 6
2.7 Fungal biomass 7
2.8 Statistical analyses 7
3 Results 8
3.1 Water chemistry and stream physical characteristics 8
3.2 Litter decomposition 9
3.3 Macroinvertebrates 10
3.4 Ergosterol 11
3.5 Nutrient content 12
3.6 Litter input 12
3.7 Variables influencing microbial decomposition 14
3.8 Macroinvertebrates 16
3.9 Estimation of litter standing stock 17
4 Discussion 18
4.1 Riparian vegetation 18
4.2 Water chemistry and stream physical characteristics 18
4.3 Adaptation to non-native species 19
4.4 Variable associations 19
4.5 Decomposer community and litter decomposition 20
4.6 Litter standing stock 21
4.7 Conclusion 21
5 Acknowledgement 22
6 References 23
Appendix 1
Appendix 2
Appendix 3

1 Introduction

1.1 Headwaters

Headwaters (catchments < 15km^2) are the dominant stream order at the landscape scale, representing up to 85% of the total stream length. Therefore, headwaters constitute a large part of the land-water interface, and act as a buffer against inputs of pollutants from the terrestrial environment for downstream lakes and streams by processing pollutants that enter headwaters (Bernhard et al. 2003, Bishop et al. 2008). Furthermore, headwaters within the same catchment can show a large variability in many important environmental factors, including water chemistry, temperature, morphology, and food resources (Temnerud and Bishop 2005, Meyer et al. 2007). This variability provides niches for a wide range of organisms that contribute to landscape-scale biodiversity (Leroy and Marks 2006). Healthy and functional headwaters are therefore important both for the function of the whole stream network and as providers of habitats to organisms. These small systems are often limited in algal production, since dense canopy cover commonly shades sunlight. With low algal production, allochthonous (from terrestrial origin) produced carbon is the dominating energy source, largely consisting of litter input from the riparian vegetation (Fisher and Likens 1973, Bärlocher 1985).

1.2 Decomposition of litter

When terrestrial litter enters the aquatic system it will start to decompose. Initially the litter will rapidly lose mass due to leaching, up to 30% within the first 24h (Petersen and Cummins 1974). Second, microorganisms, such as fungi and bacteria, colonize the litter and start to decompose its structure and change its physical and chemical characteristics (i.e. microbial conditioning). This makes the litter more palatable for detritivorous macroinvertebrates, i.e. aquatic invertebrates defined as 'shredders', which fragments the remaining litter to fine particulate organic matter (FPOM). In turn, FPOM becomes an energy source for organisms specialized to filter feeding or gathering small deposited particles (Bärlocher 1985, Webster and Benfield 1986).

The litter decomposition process is influenced by several environmental factors, and studies have tried to disentangle what these factors are. Temperature has been shown to affect decomposition mainly by increasing microorganism metabolic activity, leading to higher feeding rates, and consequently higher decomposition rates (Webster and Benfield 1986). Microbial decomposition of litter is also affected by the availability of dissolved organic carbon (DOC) in the water, as DOC is used as an energy source by bacteria. With high DOC levels, bacterial abundance increases and affects microbial decomposition positively. However, not all DOC is available for bacteria to use. Meyer et al. (1987) showed that the proportion of DOC used by bacteria changed with DOC molecule weight, indicating that not only the amount, but also the quality of DOC is important for bacterial production and consequently litter decomposition. Another important water chemistry parameter is water nutrient concentrations. Both fungal and bacterial production can be limited by nitrogen (N) and/or phosphorus (P). In comparisons of litter decomposition in nutrient rich and nutrient poor systems, a majority of studies have found higher decomposition rates in nutrient enriched systems, due to higher microbial productivity (Webster and Benfield 1986, Suberkropp and Chauvet 1995, Gulis and Suberkropp 2003, Suberkropp et al. 2010).

As already mentioned, litter decomposition is also affected by fragmentation from macroinvertebrates. However, unconditioned leaf litter contains lignin and tannins, secondary compounds that shredders lack the ability to break down. Fungi and bacteria are able to degrade secondary compounds into simpler compounds, and without this conditioning, leaf litter is a poor food resource for shredders (Graça 2001). Fungi are a more easily accessible nutrient source than leaf litter tissue to shredders, making litter colonized by microbes a preferred substrate for shredders (Bärlocher 1985). Hence, litter fragmentation by shredders is stimulated by microbial activity, and the shredder community. Although the

abundance of aquatic invertebrates affect litter decomposition rates, their community composition is also important. For example, high decomposition rates have, in several studies, been positively correlated with high species richness (Jonsson et al. 2001, Dangles and Malmqvist 2004). One plausible mechanism behind this relationship is that species have different niches, and are using litter resource slightly differently and perhaps together more efficiently (Jonsson et al. 2001).

1.3 Litter quality and stoichiometry

Although several environmental factors affect how fast leaf litter decomposes, litter quality is a strong determinant by affecting the colonization and decomposition of microbes and macroinvertebrates and subsequent decomposition rates. Studies have shown that litter decomposition is positively correlated with high N and P concentrations and negatively correlated with high lignin and tannins levels in the litter (Gessner and Chauvet 1994, Aerts 1997, Leroy and Marks 2006). High nutrient concentrations positively affect fungal and bacterial colonization, while lignin and tannins inhibit microbial growth (Webster 1986). However, quantity of nutrients in the litter is not the only factor influencing decomposition rates. Recent papers highlight the effect of litter stoichiometry, i.e. the carbon/nutrient ratios, and its effect on shredder resource consumption (Fuller et al. 2014, Manning et al. 2015). Litter of a higher carbon/nutrient ratios typically decrease consumption and growth of aquatic insects, and will therefore decompose slower than litter of low ratios. Litter stoichiometry is also affected by the amount of dissolved nutrients in the water, via impacts on fungal production. High levels of dissolved nutrients have been shown to decrease the carbon/nutrient ratio in litter via microbial nutrient uptake (Rosemond et al. 2010). Colonizing fungi can incorporate nutrients from both water and litter for production, thereby making the litter more palatable for shredders (Manning et al. 2015).

1.4 Macroinvertebrate production

As already mentioned, shredders in headwater streams rely on input of terrestrial litter as an energy source. The activity and composition of aquatic invertebrates (primarily shredders) can therefore be reflected in riparian vegetation variation. For example, a quantitative decrease in riparian vegetation input decreases production of shredders (Wallace et al. 1999). However, changes in litter quality and composition have also been found to affect secondary production in aquatic systems. Hence, streams with riparian vegetation that produce high quality litter produce more aquatic invertebrates (Kominoski et al. 2011).

Temporal variation in stream macroinvertebrate abundance can follow the abrupt increase in deciduous litter input to streams during fall, resulting in increasing growth due to increased access to food (Haapala et al. 2001). However, as decomposition of deciduous leaves is relatively fast, the majority will be consumed during fall and early winter. To avoid food shortage during and after the winter, when litter input is generally low, input of more slow-decomposing species is also important to maintain macroinvertebrate production (Cummins et al. 1989, Haapala et al. 2001). Without slow-decomposing, low-quality litter, detritivorous aquatic invertebrates will likely be limited in growth and development up until emergence in early spring to early summer, resulting in fewer macroinvertebrates that decompose litter the next fall.

1.5 Human disturbance

The variability between headwaters can partly be ascribed to disturbance in the terrestrial environment. Prior to human interference, natural disturbance such as fire, windstorms, and dying trees formed the landscape, and continuously changed the conditions for both terrestrial and aquatic systems. While dying trees only creates gaps in the forest, fire and wind disturbance can affect larger areas, even whole catchments (Essen et al. 1997). However, today almost 97% of the productive forested area in Sweden is affected by forestry (Kempe 2014), which is the single most common disturbance in Swedish forests (Essen et al. 1997). To replace fossil energy sources, the demand to use forest products is increasing. In

Sweden, one attempt to increase the production of forest products was the introduction of the North American tree species lodgepole pine (*Pinus contorta*). Today it occurs mainly in the northern parts of Sweden where it covers approximately 600,000 ha, i.e. 4% of the productive forested area. Lodgepole pine produces more litter but of lower quality, compared to the native Scots pine (Engelmark et al. 2001). For aquatic ecosystems, the effects from forestry practices can be seen in changed abiotic factors, such as increased temperature, reduced canopy cover, higher stream flow, and changes in water chemistry (Allan 2004). Harvesting of the riparian vegetation can also decrease litter input to the aquatic system (Webster et al. 1992) and result in changed tree species composition. The present management action to prevent damage from forestry on aquatic ecosystem is to create riparian buffer zones along streams and rivers to maintain processes in the riparian zone, including filtration of pollutants, sediment stabilization, shading, and water regulation, but this is not practiced along headwaters (Kuglerova et al. 2014).

1.6 Study aims

Headwaters receive relatively little focus despite their contribution to stream networks and importance for downstream aquatic systems (Bishop et al. 2008). Litter decomposition processes in headwaters are important for both production of aquatic invertebrates and for the uptake and recycling of nutrients and carbon in forests. Human disturbance transforms landscapes, and it is increasingly important to understand how this affects vital ecosystem processes. The majority of studies on decomposition in headwaters compares relatively few locations (Gessner and Chauvet 1994, Suberkropp and Chauvet 1995, Haapala et al. 2001, Lecerf et al. 2007), despite the innate heterogeneity of these systems, and seldom tries to include all parameters that are known to affect litter decomposition. This study tries to disentangle (I) which variables are important for decomposition of four leaf litter species, using a wide range of sites and variables. Three of the species; alder (*Alnus glatinosa*), birch (*Betula pubecens*) and spruce (*Picea abies*), are native in the area but differ in quality, whereas the fourth species, lodgepole pine, is alien to the studied systems. Using both native and non-native species makes it possible to also investigate (II) if decomposer in headwater systems are adapted to current condition or can adapt to sudden change in litter input. Comparing microbial- and macroinvertebrate-mediated litter decomposition makes it possible to assess (III) the relative importance of microbes and macroinvertebrates for litter decomposition in headwater streams. Lastly, the collected data allows for estimations of leaf litter quantities in the studied headwaters to (IV) detect risks of food limitations for macroinvertebrates over the studied season.

2 Material and methods

2.1 Study site

The study was conducted in 20 first- to second-order streams in the Swedish province of Västerbotten and Ångermanland (Figure 1). Mixed boreal coniferous forest is the main forest type within the study area. The streambeds varied from being dominated by gravel and sand to largely consisting of smaller boulders covered with aquatic mosses. Forestry practice is the major disturbance factor in the area, with clear cutting followed by soil scarification and plantation of coniferous trees as the main method (Essen et al. 1997). The studied streams differ in type of land cover and surrounding forest structure (i.e. forest age and species composition), from young forest dominated by alder and birch to old spruce forests, creating a gradient in external influences on the headwater sites. The study was carried out in 2014, from May to November. At each stream, 100 m that was dominated by riffles was chosen as study site.

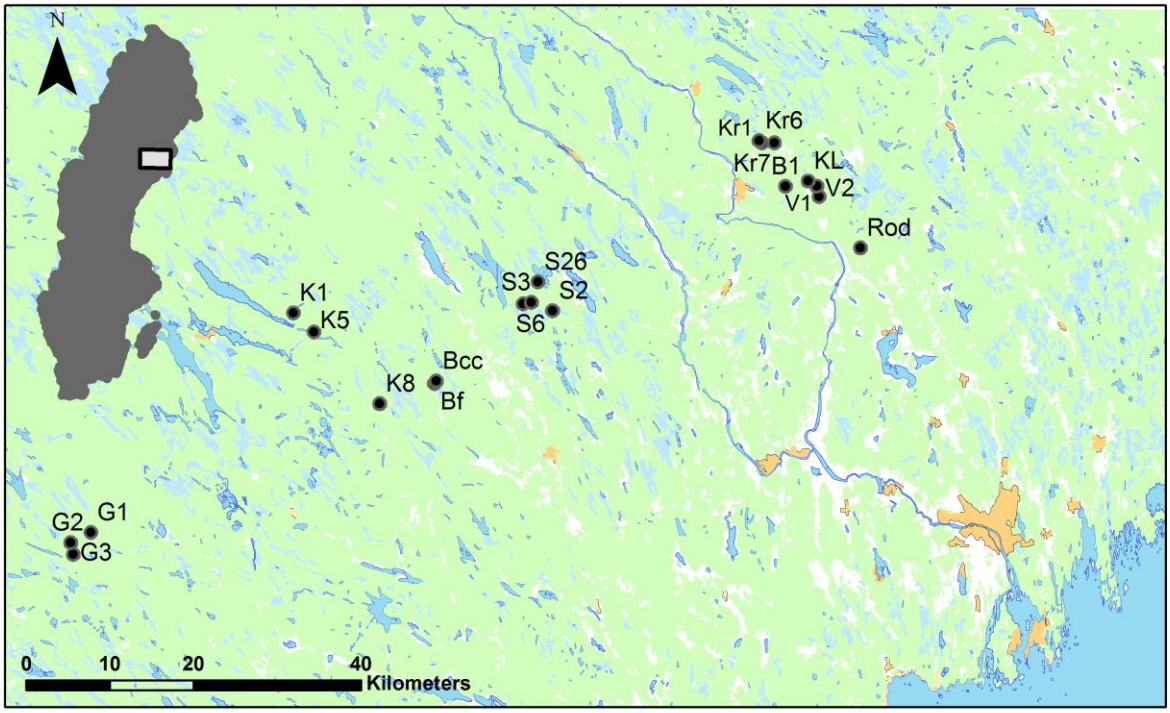

Figure 1. Map of study area where each point represents a site.

2.2 Water chemistry and stream characteristics

Water samples were taken four times during August to November for measures of absorbance, DOC, reactive phosphorous (PO_4), nitrate (NO_3), ammonium (NH_4), dissolved organic nitrogen (DIN), total nitrogen (TN), conductivity and pH. Samples where filtered in the field with a 0.45-µm nylon membrane filter (Sarstedt, Nümbrecht, Germany). All samples were stored during fieldwork in a cooling box with ice packs. In the lab, pH and absorbance samples were stored at +4 ºC. DOC and nutrient samples were stored at -25 ºC before being analyzed. All water chemistry samples were analyzed at the Swedish University of Agricultural Science (SLU) in Umeå. From the absorbance spectra SUVA254 was estimated. SUVA254 is the UV absorbance at wavelength 254 and is used to estimate the content of aromatic carbon (Weishaar et al. 2003). Compared to other carbon molecules, aromatic carbons are less available for organisms to use (Perdue 1998). At the first water sampling occasion, not enough water volume was collected for the SUVA254 measurements. Only three replicates for SUVA254 were therefore used. Water temperature was continuously monitored during the experiment with HOBO® Pendant temp/light loggers that were attached to an iron bar at each site. On occasion, water levels were below the logger, but those readings were brief and could be removed from the temperature data. In total, 64 days were included, after an exclusion of 23 days for all sites. Light condition was determined in August before leaf abscission using a spherical densiometer. Stream depth, width and velocity were measured at three points along the study site in each stream. For depth, the measuring was repeated three times during the study period to get an average for the whole period.

2.3 Litter decomposition

2.3.1 Study design

Lodgepole pine and Norway spruce needles where collected in early June by shaking branches, which makes dry needles come off. Alder and birch leaves were collected at abscission. The needles and leaves were air-dried indoors (20 °C) to constant weight.

To measure microbial decomposition 15x15-cm fine-mesh (0.5 mm) bags were used to exclude effects from invertebrates. Another set of 12x17-cm coarse-mesh (5 mm) bags was used to measure invertebrate leaf litter decomposition. In each bag, 3.0 (±0.05) g of air-dried leaves from one of the studied species was inserted. For the coarse mesh bags, only birch leaves were used. The stalks on birch and alder leaves were removed before weighing and insertion into the bags. Unlike the fine mesh bags, the coarse mesh bags was not closed by folding the top over and secured with staples, but by pressing the sides together and closed with plastic strings to create more space in the bag.

At each stream site, five replicates of each litter species and five coarse-mesh bags containing birch (25 in total) were anchored randomly to a chain secured with an iron bar in the streambed. The bags were placed some distance from each other to prevent overlapping. Bags containing coniferous species where introduced to the streams in August (week 34) and bags with deciduous species in September (week 39). The longer time in the streams for conifers is due to their known slower decomposition rates. Therefore, more time is needed to see potential difference among sites. The litterbags were retrieved in November (week 47), rendering a total of 87-91 days for coniferous litter and 53-56 days for deciduous litter. Before processing, the bags were stored at -18 °C. This study design that performed one litterbag collection, as opposed to sequential collections, was chosen to be able to include a high number of sites in the experiment. With more sites, a wider range of variation in the variables presumed to influence litter decomposition can be obtained.

2.3.2 Mass loss

The decomposed litter was rinsed before being oven dried at 60 °C for 48h. The dried litter was then combusted in 550 °C for 40 min to obtain ash weight (Benfield 1996). Litter ash-free dry mass (AFDM), used in the statistical analyses, was obtained by subtracting ash weight from dry-mass.

Following the same procedure as above, initial AFDM was obtained by combusting three 1 g samples of each of the four litter species. The average from the tree samples was used to calculate initial AFDM. Mass loss was then calculated by subtracting AFDM remaining from initial AFDM (Eq. 1):

$$M_{Loss}=AFDM_{Initial}-AFDM_{Remaining} \quad (1)$$

2.3.3 Decomposition rate

To calculate decomposition rate for each treatment and site, the decomposition constant (k) was calculated, using the negative exponential decay model (Eq. 2):

$$k= \ln(M_t/M_o)/t \quad (2)$$

where M_t is the AFDM at time t and M_o is the initial AFDM. A decision was made to use the same time of exposure for each species despite a maximum difference of three days between sites. This was because when comparing the results of using different exposure times, a difference in two days had a greater impact on k than two days longer in the field would have had on litter mass loss since decomposition rate slows down with time as water-soluble compounds and easily decomposed material is lost early in decomposition processes (Webster and Benfield 1986).

There was a significant positive correlation between mass loss and *k* (Pearson's product-moment correlation, $p<0.001$, $r=0.94$). Hence, in all analyses, *k* instead of mass loss was used to enable comparison between species with different time in the field

2.3.4 Nutrient content

The initial litter quality of each species was measured by sending litter to SLU Umeå for analyses of C and N content. The C/N ratio was used as a proxy for litter quality.

2.4 Litter input

To quantify amount and type of litter input to each stream, three litter traps, i.e. trays that were 40x50 cm in size, were placed immediately adjacent to each stream channel at each study site in late May. To be able to assess potentially different effects of summer and fall litter input, trapped litter was collected at the end of summer (20-22 August) and in late fall (late October), the latter date being after complete defoliation of deciduous trees. The collected litter was sorted to six categories (i.e. birch leaf, alder leaf, pine needle, spruce needle, twig, and 'other') and was oven dried at 60 °C for 48h to obtain dry mass. The category 'other' consisted mainly of grasses and herbs. The dry litter was then combusted to obtain AFDM as above.

2.5 Estimation of litter standing stock

Calculations on litter standing stock were made to assess if shredders could be limited in growth by lack of food at the end of the study. However, since both litter input and decomposition data only was available for alder, birch and spruce, estimations were only done for those species. Incoming litter to the stream, based on the litter trap data, was set to decompose over 53 days according to the observed decomposition rates from the litterbag experiment. Only litter input from the fall period was used, since decomposition rates were based on that time period. I assumed that fall litter input followed the same temporal pattern across all sites, and neglected downstream litter transport, as it likely would be compensated for by upstream transport into the study site. Therefore, in the calculations, all litter was entering the stream at the same time and no net loss of standing stock due to transport was assumed. For Birch, the effect of macroinvertebrate feeding on litter standing stock could also be estimated. The calculations were based on the negative exponential decay model, where each species was calculated separately (Eq. 3) and the summarized (Eq. 4):

$$M_{53}=M_o*e^{-kt} \quad (3)$$

$$M_{tot}=M_{Alder}+M_{Birch}+M_{Spruce} \quad (4)$$

where M_t is the litter mass remaining at time t and *k* is the decomposition rate. M_{Alder} is the mass remaining for alder at time t, and the same for M_{Birch} and M_{Spruce}.

2.6 Macroinvertebrates

The coarse-mesh bags were retrieved at the same time as the fine mesh bags using a hand net to prevent loss of invertebrates when removing the bags. By rinsing the leaves over a 35-mm mesh sieve, the invertebrates were removed, picked and preserved in 70% ethanol before being sent to a taxonomist for identification to species/genera. The species were then categorized to functional feeding groups using previous classification of the species (Schmidt-Kloiber and Hering 2012). In later analyses, only shredders were considered, whereas other functional feeding groups were not used in the study. Shredder invertebrates were dried at 60 °C for 48 h to estimate dry weight.

2.7 Fungal biomass

In the laboratory, samples of approximate 30 mg of litter were taken from the fine-mesh bags. The samples where frozen and stored at -18 ºC, before being analyzed for ergosterol content at SLU Uppsala. Measuring ergosterol levels is a common method for estimation of fungal biomass, since ergosterol is found exclusively in cell membrane of living fungi (Gessner and Chauvet 1994). Since leaf mass was taken from the litterbags for the ergosterol analyses before they were processed and weighed, the proportion of AFDM of the remaining litter was multiplied with the dry weight from the ergosterol sample and added to the AFDM of the fine-mesh bag it originated from.

2.8 Statistical analyses

Differences in decomposition were analyzed using one-way analysis of variance (ANOVA), with litter species as a fixed factor and decomposition rate as the response variable. Tukey's HSD for pair-wise comparisons was used to determine significant difference among litter species. The same analysis was performed with ergosterol content as response variable. Two two-way ANOVAs were conducted to test for significant differences in litter input between the summer and fall period, with time period and litter categories as fixed factors and the different categories proportional or biomass input as response variable. Invertebrate effects on decomposition were determent in a paired t-test, comparing decomposition rates for birch in fine- and coarse-mesh litterbags. To assess if all sites followed the same pattern of differences in decomposition rates between fine- and course-mesh bags, a two-way ANOVA was conducted, with site and treatment as fixed factors and decomposition rate as response variable, followed by Tukey's HSD.

To explore relationships between abiotic factors and riparian vegetation at each stream site, a principal component analysis (PCA) was conducted. Two PCAs, one for summer and one for fall, were also conducted to visually describe differences in the composition of litter input between streams. Component 1 scores from these two PCAs were later used as single measures of litter input composition. Partial least squared (PLS) models were used to determine which variables were the most influential for determining decomposition for each litter species. For the coarse-mesh bags, invertebrate variables were also included (Table 1). A PLS analysis is comparable to a multiple regression analysis, but is preferred when dealing with large numbers of independent variables and relatively low numbers of observations (i.e. sites). Additionally, a PLS does not assume normally distributed data, which in field studies often is difficult to obtain (Hulland et al 2010). The evaluation of the PLS models was based on the level of variance explained (R^2), loadings of the independent variables, and the variable influence on projection (VIP). The independent variable loading describes the relative strength and direction of the relationship between independent and response variable. The VIP value summarizes the importance of each variable (Stenroth et al. 2015). In the models, the limit for a variable to be included in the final model was a VIP value at 0.7. To visually describe difference in variable relationships between litter species, a PCA bi-plot was conducted on the fine-mesh bags and the most important variables from the PLS analyses.

Litter availability for shredders can either depend on amount of litter input from the riparian vegetation or the decomposition rate of standing stock. Hence, two PLS models were conducted to assess the relative importance of litter input and decomposition rate for food availability to shredders. The first model included abiotic variables combined with microbial decomposition and litter input of alder, birch, and spruce. The second model also included macroinvertebrate decomposition, but was only done for birch litter. All analyses were performed in version 2.15.1 of R (R Core Team 2012).

Table 1. Variables used in the PLS models on litter decomposition. For litter input, summer and fall values were kept separate. For the ergosterol variable, only ergosterol for the analyzed species was used.

Variables	Analyze used
Litter input	All
Alder	“
Birch	“
Pine	“
Spruce	“
Twigs	“
Other	“
Total litter input	“
Water chemistry	
Conductivity	“
pH	“
DOC	“
SUVA254	“
NH_4	“
NO_3	“
DIN	“
TN	“
PO_4	“
Stream characteristics	
Water depth	“
Chanel width	“
Stream velocity	“
Canopy openness	“
Temperature	“
Invertebrates	
Species richness	Coarse
No. individuals	“
Dry weight	“
Other	
Ergosterol	All
Litter input summer PCA	“
Litter input fall PCA	“

3 Results

3.1 Water chemistry and stream physical characteristics

Water chemistry variables varied among sites and over the study period (Appendix 1). Mean temperature during the study ranged from 3.81 °C at site G1 to 4.85 °C at site G2 (Table 2). Site Bcc stood out in canopy cover with a canopy openness of 87.4%, while the other sites ranged from 1.4 to 22.1%. Both stream channel width and mean depth was highly variable among sites. The widest channel was found at site V1 (185.0 cm), while the deepest site was site KL with mean depth of 26.6 cm. Site B1 had the lowest stream flow at 0.025 m/s, while the highest velocity was found at site BF (0.307m/s). All site characteristics can be viewed in table 2.

Table 2. Mean values for stream physical characteristics for each site. Values in parentheses show ± 1 standard error.

Site	Depth (cm)		Width (cm)		Velocity (m/s)		Temperature (°C)		Canopy openness (%)	
B1	22.8	(1.9)	121.8	(18.3)	0.025	(0.011)	4.42	(0.055)	5.72	(2.08)
Bcc	7	(0.4)	43.8	(5.8)	0.126	(0.032)	4.36	(0.045)	87.36	(8.61)
BF	13.4	(0.6)	101.7	(8.6)	0.307	(0.060)	4.35	(0.051)	9.88	(0.69)
G1	16.3	(4.4)	81.7	(7.0)	0.133	(0.038)	3.81	(0.049)	22.1	(10.95)
G2	9.4	(0.7)	83.2	(7.4)	0.181	(0.049)	4.85	(0.072)	6.15	(1.25)
G3	20.8	(2.8)	73.3	(7.2)	0.114	(0.038)	4	(0.054)	15.43	(4.21)
K1	8.3	(0.7)	39.7	(3.9)	0.276	(0.053)	4.23	(0.032)	11.53	(2.65)
K5	10.1	(0.8)	65.3	(8.8)	0.074	(0.025)	4.65	(0.049)	16.55	(2.62)
K8	10.1	(0.9)	67.5	(9.1)	0.229	(0.037)	4.44	(0.053)	1.47	(1.09)
KL	26.6	(3.4)	144.3	(21.8)	0.25	(0.035)	4.13	(0.055)	3.21	(2.30)
Kr1	6.2	(0.6)	59	(6.9)	0.097	(0.016)	4.25	(0.048)	3.64	(1.48)
Kr6	9.9	(1.3)	84.7	(20.2)	0.101	(0.047)	4.5	(0.064)	8.58	(3.90)
Kr7	6.9	(0.5)	85.2	(9.7)	0.287	(0.184)	4.33	(0.048)	5.2	(2.80)
Rod	16.5	(1.9)	139.2	(4.2)	0.227	(0.054)	4.04	(0.059)	8.93	(5.04)
S2	9.4	(0.9)	40.7	(2.9)	0.065	(0.019)	4.37	(0.052)	1.39	(0.38)
S26	12.1	(0.5)	50.3	(5.3)	0.102	(0.026)	3.93	(0.056)	9.36	(1.30)
S3	5.7	(1.2)	73.8	(3.1)	0.047	(0.027)	3.87	(0.053)	1.82	(0.60)
S6	5.8	(0.5)	56.2	(8.9)	0.065	(0.009)	4.17	(0.057)	3.81	(1.96)
V1	14.3	(1.4)	185	(6.3)	0.233	(0.028)	4	(0.054)	15.34	(10.68)
V2	20.3	(1.6)	104.2	(10.2)	0.298	(0.085)	4	(0.056)	5.55	(2.69)

3.2 Litter decomposition

There was a significant difference in decomposition rate for fine-mesh bags between litter species (df=3, F=227, p<0.001). Tukey's HSD showed that there was significant difference between all species (Figure 2). Alder had the highest mean decomposition rate (*k*=0.0077) and lodge pine the lowest (*k*=0.0025).

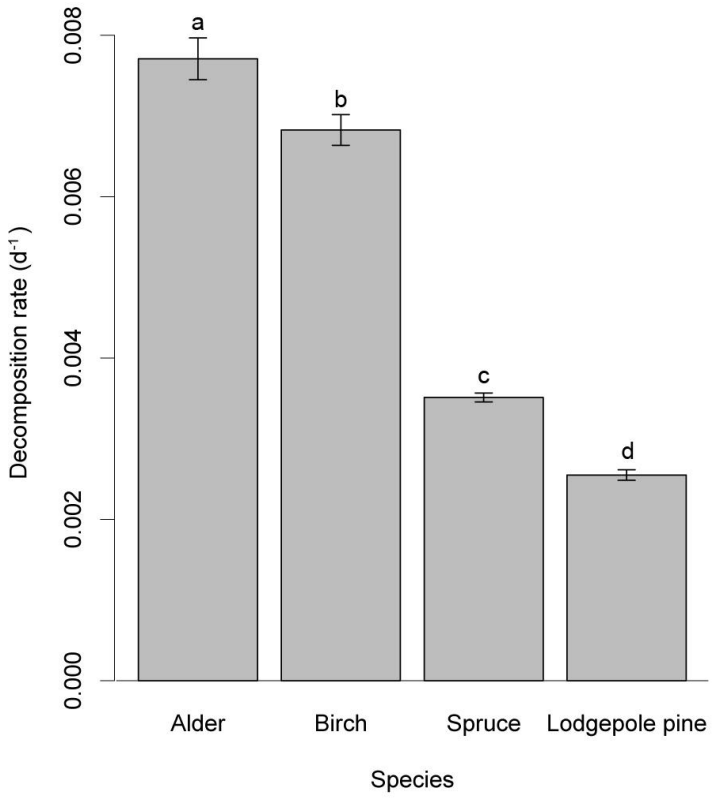

Figure 2. Decomposition rate (d^{-1}) for studied litter species. Different letter indicate significant difference at p=0.05. Error bars indicate ± 1 SE.

Decomposition rates for the coarse-mesh bags ranged between 0.009-0.024 d^{-1}, with highest decomposition at site V1 and the lowest at site G3 (Table 3). In general, coarse-mesh bags (i.e. shredders present) showed a higher average birch decomposition rate than fine-mesh bags (i.e. microbes) (df = 19, t = -10.0946, p<0.01). However, within sites only 11 of the 20 sites showed a significant higher birch decomposition rate due to shredder presence (Table 2).

Table 3. Mean decomposition rate of birch in fine-mesh (Microbes) and coarse-mesh (Microbes+Shredders) bags at each stream. Values in parentheses show ± 1 standard error. *p* shows the result from a two-way ANOVA comparing decomposition of birch with and without influence of shredders. Significant differences (p<0.05) are bolded, and close to significant difference are presented in cursive.

	Decomposition rate (d^{-1})				
Site	Microbes		Microbes+Shredders		p
B1	0.0055	(0.0002)	0.013	(0.002)	**0.006**
Bcc	0.0087	(0.0008)	0.021	(0.001)	**< 0.001**
BF	0.0068	(0.0003)	0.017	(0.001)	**< 0.001**
G1	0.0067	(0.0003)	0.015	(0.002)	**< 0.001**
G2	0.0070	(0.0002)	0.013	(0.002)	*0.062*
G3	0.0066	(0.0008)	0.009	(0.001)	0.999
K1	0.0081	(0.0008)	0.016	(0.001)	**0.004**
K5	0.0054	(0.0003)	0.012	(0.002)	**0.037**
K8	0.0068	(0.0003)	0.012	(0.001)	0.3
KL	0.0073	(0.0005)	0.017	(0.001)	**< 0.001**
Kr1	0.0061	(0.0004)	0.011	(0.001)	0.308
Kr6	0.0065	(0.0004)	0.014	(0.001)	**0.002**
Kr7	0.0079	(0.0004)	0.012	(0.001)	0.764
Rod	0.0066	(0.0003)	0.012	(0.001)	0.357
S2	0.0070	(0.0005)	0.011	(0.001)	0.894
S26	0.0064	(0.0001)	0.018	(0.001)	**< 0.001**
S3	0.0058	(0.0004)	0.010	(0.002)	0.793
S6	0.0069	(0.0004)	0.017	(0.001)	**< 0.001**
V1	0.0079	(0.0007)	0.024	(0.004)	**< 0.001**
V2	0.0066	(0.0005)	0.013	(0.001)	*0.085*

3.3 Macroinvertebrates

In total, 20 shredder species were found in the coarse-mesh bags (Appendix 2). Mean species richness ranged from 0.6 to 7.6 species per litterbag and was highest at site Rod (Table 4). Mean abundance was highly variable, ranging from 6 at site G3 to 113.2 at site Bcc. Mean macroinvertebrate biomass (dry weight) ranged from 1.2 to 16.3 mg.

Table 4. Mean invertebrate species richness, abundance, and biomass from the coarse-mesh bags. Values in parentheses show ± 1 standard error.

Site	Species richness		Abundance (ind/bag)		Dry weight (mg)	
B1	2.2	(0.4)	63.4	(16.1)	4.2	(1.64)
Bcc	3	(0.3)	113.2	(26.2)	13.1	(2.61)
BF	4.6	(0.5)	22.4	(2.4)	4.1	(0.25)
G1	2.8	(0.4)	24	(5.9)	1.2	(0.56)
G2	4.6	(0.7)	16.8	(4.7)	3.5	(0.42)
G3	2.4	(0.5)	6	(1.4)	2	(0.11)
K1	4.8	(0.6)	79.6	(15.0)	4.2	(1.49)
K5	2.2	(0.4)	9.4	(2.2)	4.3	(0.19)
K8	3.2	(1.0)	30.2	(9.3)	3.7	(0.85)
KL	5.6	(0.6)	47	(17.7)	7.4	(1.77)
Kr1	3.2	(0.4)	26.2	(5.0)	2.3	(0.49)
Kr6	2.8	(0.4)	55.8	(22.5)	5.4	(2.23)
Kr7	1.6	(0.5)	26.8	(8.2)	1.9	(0.79)
Rod	7.6	(0.4)	33.8	(2.4)	11.7	(0.25)
S2	3	(0.3)	103.8	(30.0)	7.3	(2.98)
S26	2.6	(0.2)	7	(1.3)	16.3	(0.14)
S3	0.6	(0.2)	35.8	(16.0)	2.1	(1.58)
S6	1.8	(0.2)	46	(10.2)	9.2	(1.04)
V1	4.2	(1.1)	15.8	(3.5)	3.5	(0.29)
V2	4	(0.5)	14	(3.8)	6.1	(0.34)

3.4 Ergosterol

There was a significant difference in ergosterol content among litter species (df=3, F=41.18, p< 0.001). However, Tukey's HSD analysis revealed that only birch was significantly different from the other species (Figure 3).

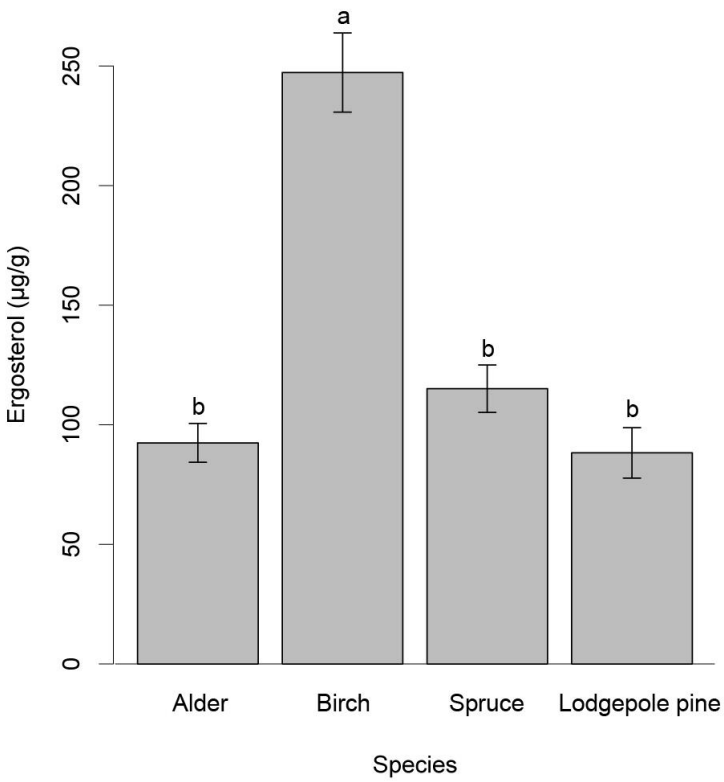

Figure 3. Mean ergosterol content in different leaf litter species after the end of the study. Different letters indicate significant differences. Error bars represent ± 1 SE.

3.5 Nutrient content

The analyses of initial nutrient content showed that alder contained the lowest C/N ratio (16.6), followed by birch (36.2), spruce (45.1) and lodgepole pine with the highest (51.3).

3.6 Litter input

Litter input varied among sites (Appendix 3), and was significantly higher in fall (t-test, df = 19, p< 0.001). The Tukey's HDS showed that, in fall, the proportion of birch input was significantly higher (p< 0.001), and the proportion of pine (p< 0.001) and spruce (p<0.001) was significantly lower than during the summer (Figure 4a). Total birch biomass input was also significantly higher in fall compared to in summer (p< 0.001) (Figure 4b). Since biomass does not describe riparian vegetation input, the proportion of total input was used in the analyses.

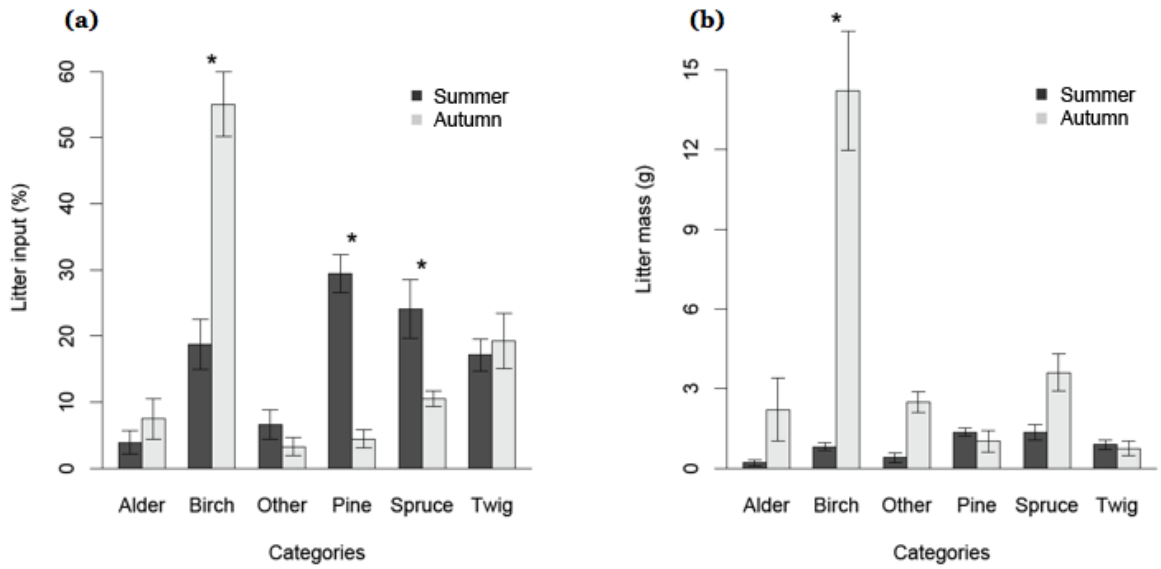

Figure 4. Difference in (a) percent of total and (b) total biomass input for different litter categories between summer and fall. * indicates significant differences between the periods at p = 0.05. Errors bars represent ± 1 SE

Both PCAs for litter input showed negative relationships between amount of spruce needles and birch litter (Figure 5). The same negative relationship was seen between alder and pine. For litter input composition in summer, positive component 1 scores represent domination of spruce and negative values represent domination of birch. Litter input composition during fall showed the opposite pattern.

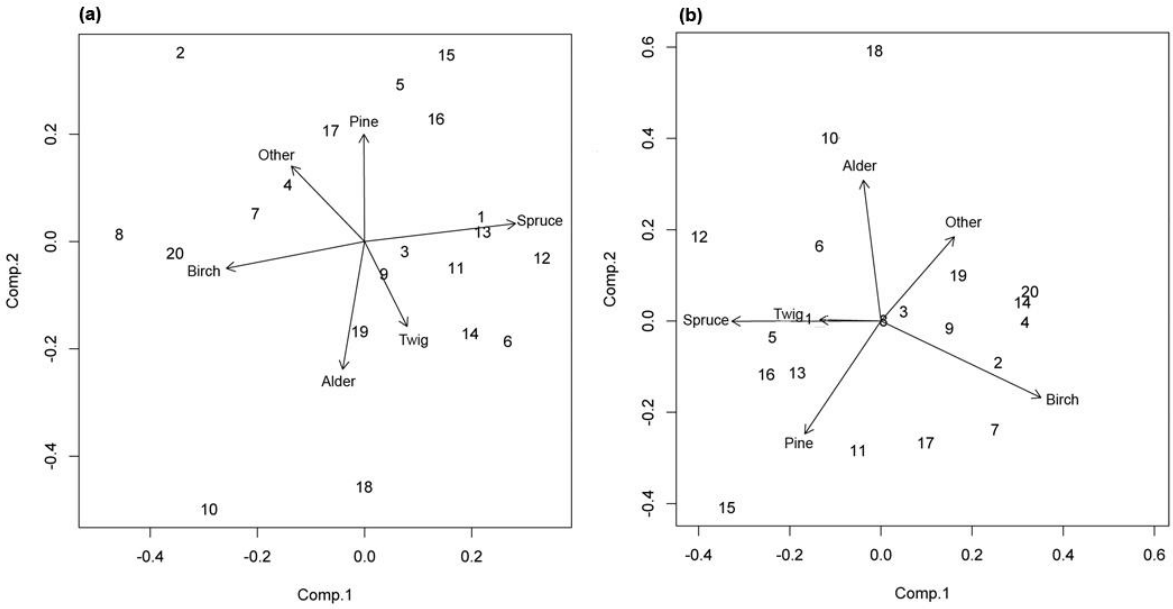

Figure 5. Principal component analysis (PCA) bi-plot of litter input composition during (a) summer and (b) fall.

The PCA on abiotic factors and riparian vegetation input showed that sites with high input of birch were associated with high SUVA254 values, high input of other vegetation than needles and leaves (i.e. 'other'), high canopy openness, and high stream velocity (Figure 6). Spruce input was negatively correlated with birch input, but positively related to twig and pine input, conductivity, and total litter input during summer. Alder showed a positive correlation with total input during fall and negative correlation with canopy openness, NO_3, and DIN.

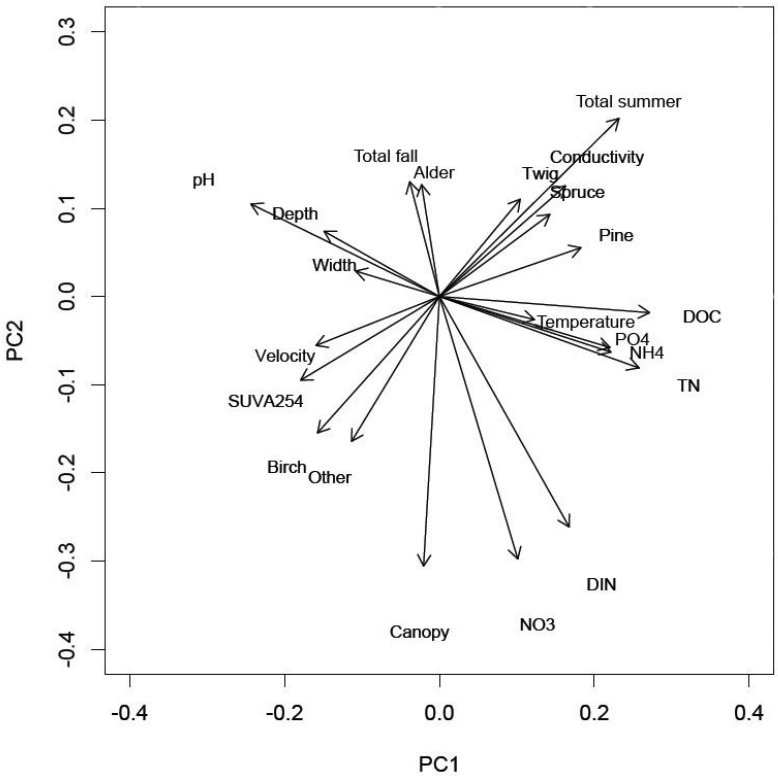

Figure 6. Principal component analysis (PCA) bi-plot of environmental variables, including litter input from alder, birch, pine, spruce twig and other materials. PC1 and PC2 together explained 40.8% of the variation (23.5% and 17.24%, respectively).

3.7 Variables influencing microbial decomposition

The PLS model for alder decomposition identified stream velocity as the most important variable, with a positive relationship (Figure 7a). Other important variables were total litter input during summer, canopy openness, SUVA254, and proportion of 'other' in summer litter input. Of the five most important variables, only total litter input during summer was negatively correlated to the decomposition of alder. For birch decomposition, proportion of 'other' in the summer litter input was the most important variable, followed by canopy openness, levels of NO_3 in the water, litter input composition in fall, and amount of twigs in litter input during fall (Figure 7b). The positive correlation with litter input composition means that decomposition of birch litter is faster at sites where birch dominates fall input. Only proportion of twigs in litter input during fall was negatively related to decomposition. Litter input composition during summer was the most important variable for explaining decomposition of lodgepole pine. However, also proportion of spruce in both summer and fall litter input explained large amounts of variation in lodgepole pine decomposition. Not surprisingly, amount of birch in summer litter input was negatively correlated with lodgepole pine decomposition (Figure 7c). For spruce decomposition, DOC was the most important variable, with a negative relationship (Figure 7d). Also total litter input during summer and TN concentration showed negative correlations to spruce decomposition, whereas there were positive relationships with litter input composition in fall (i.e. domination of birch in litter input) and pH. Because of higher decomposition rate, alder and birch decomposition had greater impact on total litter mass loss than lodgepole pine and spruce. The PLS for total mass loss was therefore influenced by the same variables as decomposition of alder and birch (data not shown).

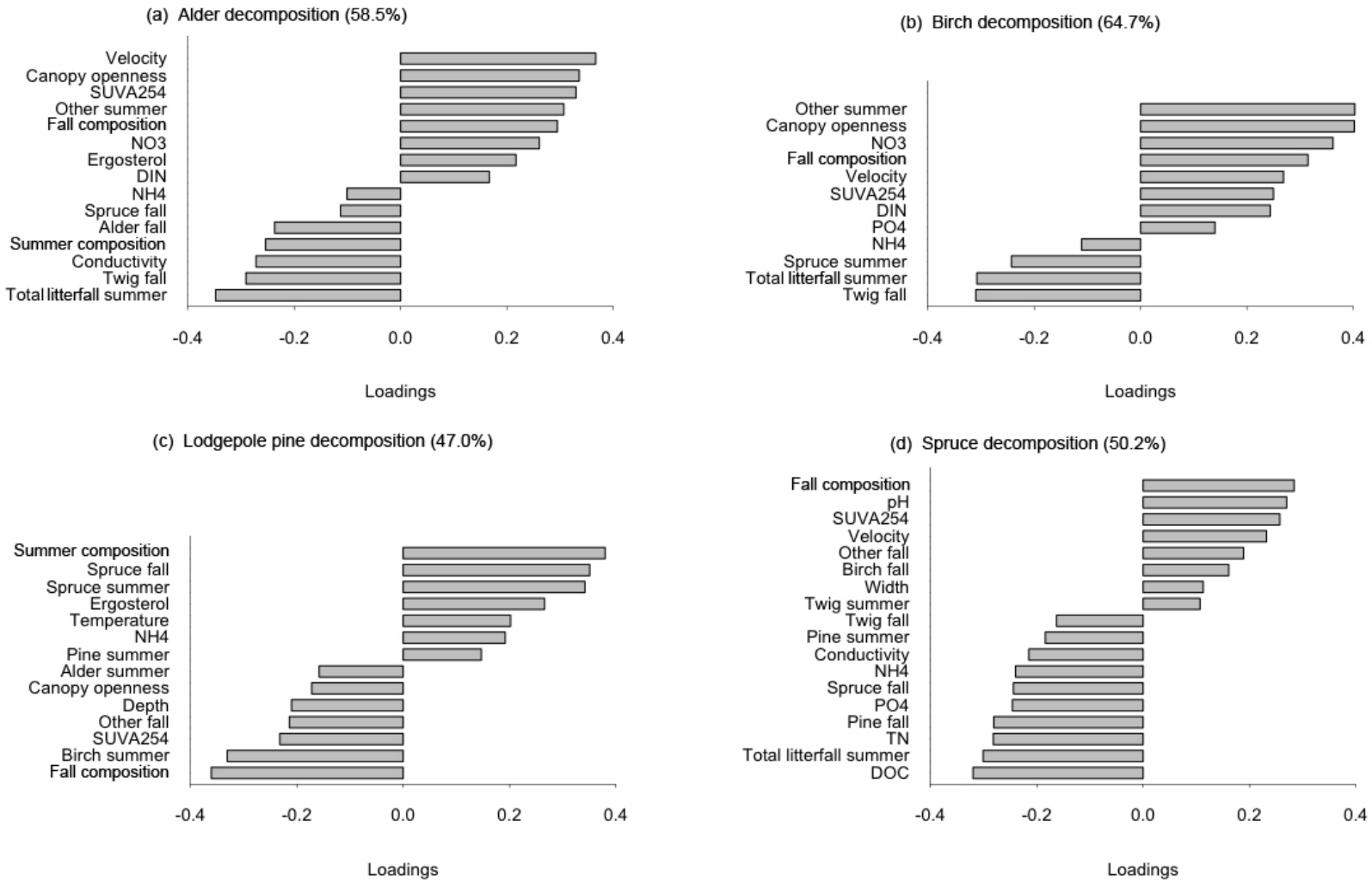

Figure 7. Results from partial least square (PLS) models on microbial decomposition of (a) alder, (b) birch, (c) lodgepole pine, and (d) spruce. The loading value is the strength of each variable on the model and positive/negative values indicate direction of influence. The percentage in the parentheses is percent variation explained by the first component of the model.

The PCA on explanatory variables and decomposition rates of different litter species showed that decomposition rates of the deciduous species, i.e. alder and birch, were positively related to each other (Figure 7). Decomposition rates of conifers were however not correlated, and only lodgepole pine showed a negative relationship with decomposition of both alder and birch.

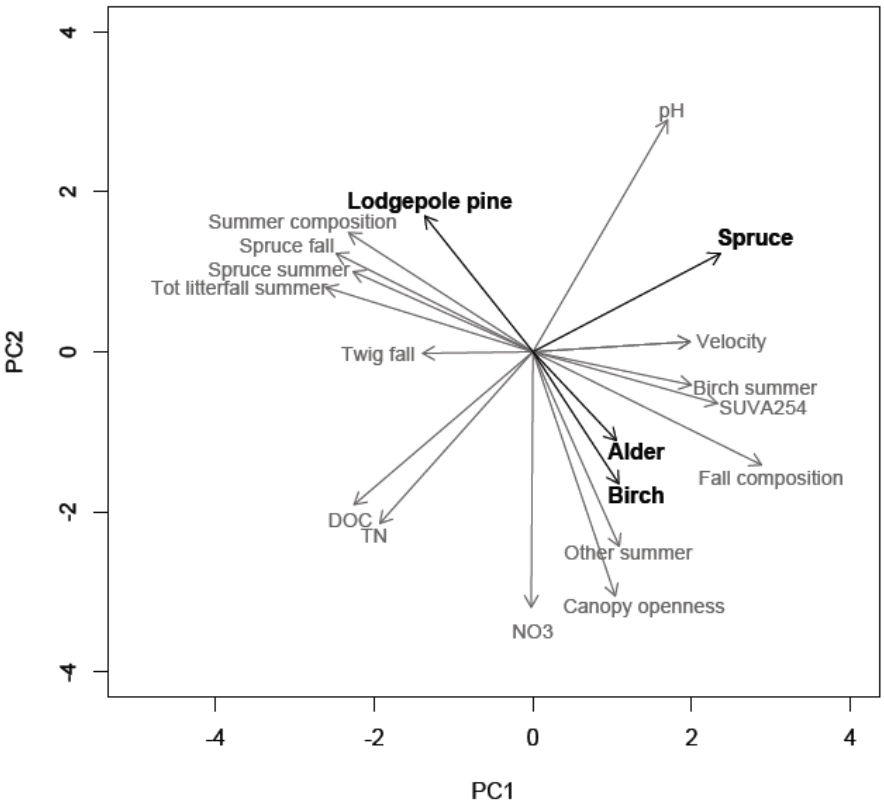

Figure 8. Principal component analysis (PCA) bi-plot showing relationships between explanatory variables (grey) and microbial litter decomposition rates for the four litter species (bold). Only the five most influential variables from the PLS models (Figure 6a-d) are included. PC1 and PC2 axis together explained 34.18% of the variance in the data (19.64% and 14.52%, respectively).

3.8 Macroinvertebrates

Invertebrate species richness and biomass (both positive) were less important for birch litter decomposition than total summer litter input and litter input composition in summer (both negative). Also, canopy openness and litter input composition in fall, both with positive relation to decomposition, showed higher influence (Figure 9). Since input of spruce needles dominates litter input during summer, the negative relation to total summer input and litter input composition in summer means that decomposition is lower where spruce dominates the riparian vegetation. Positive relation to litter input composition in fall means that decomposition is higher at sites where birch leaves dominate litter input in fall.

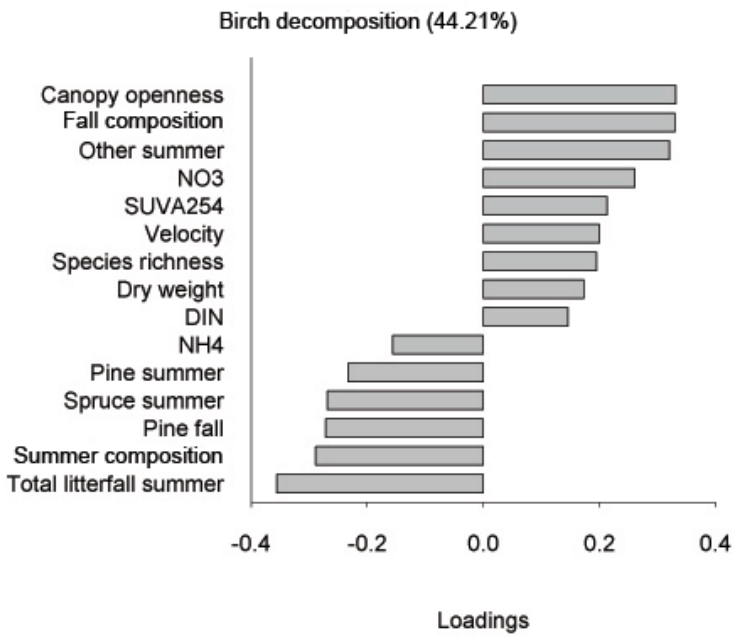

Figure 9. Result from PLS analysis on birch decomposition by both microbes and shredders. The loading value is the strength of each variable on the model and positive/negative values indicate direction of influence. The percentage in parentheses is the variance explained by the first component of the PLS.

3.9 Estimation of litter standing stock

Litter loss of standing stock (i.e. alder, birch and spruce litter input in fall) due to microbial decomposition (and leaching) ranged from 18.6% at site Kr1 to 36.3% at site Bcc. For birch litter decomposition, an inclusion of shredder fragmentation in the calculations significantly increased mass loss (t-test, df = 19, t = 6.79, p <0.001) with mass loss ranging from 39% to 72%. The PLS models showed that amount of litter input was the most influential variable for predicting availability of litter standing stock at the end of the study period (Figure 9a). When including macroinvertebrate consumption in the analysis, decomposition rate was also an important determinant (Figure 9b). However, results from both analyses showed that litter-standing stock primarily is determined by litter input and less so by microbial and shredder decomposition rates.

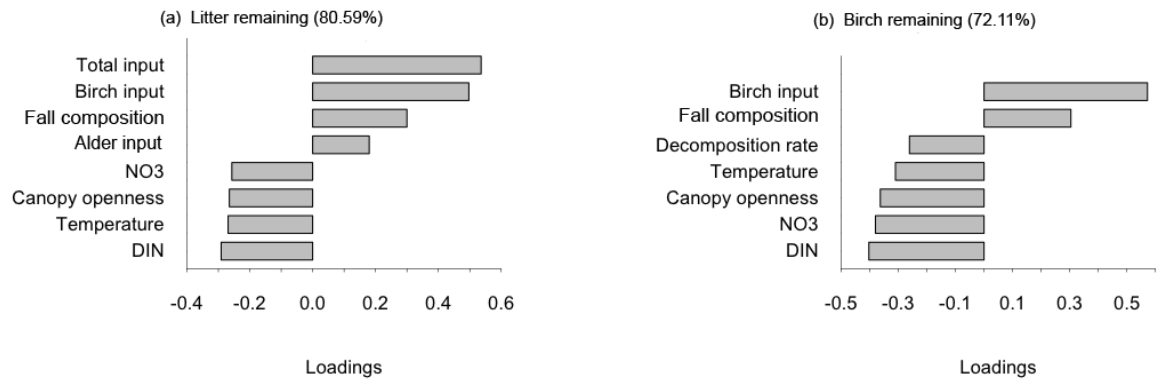

Figure 10. Result from PLS analyses on estimation of litter standing stock after the study period and the influence of different variables. (a) includes litter input from alder, birch and spruce, and only considers microbial decomposition, whereas (b) only includes input from birch but includes macroinvertebrate consumption as well. The loading value is the strength of each variable on the model and positive/negative values indicate direction of influence. The percentage in parentheses is the variance explained by the first component of the PLS.

4 Discussion

Previous studies have shown that several factors, such as nutrient availability and temperature, affect litter decomposition (Webster and Benfield 1986). My results are similar to those previously found, but I also found that the effect of these factors are much more complex because litter decomposition of different species were affected differently. Additionally, I found a high influence of riparian vegetation composition on the in-stream litter decomposition processes.

4.1 Riparian vegetation

The results from this study show that variation in riparian vegetation, and thus the quantity and quality of litter input to streams, can have large effects on the in-stream litter decomposition processes (i.e. carbon and nutrient cycling). Such an influence has only before been shown for decomposition of high-quality litter (e.g. Kominoski et al. 2011), while I found that all types of leaf litter, including poor-quality litter, were affected. A possible explanation for increased litter decomposition in streams surrounded by high-quality litter species – in this case birch – is that these streams have a greater microbial biomass (Webster 1986), leading to higher decomposition of both high- and low-quality leaf litter. However, this argument does not hold for lodgepole pine needles that were of lowest quality (and exhibited the overall slowest decomposition) but decomposed the fastest in streams where poor-quality litter (i.e. spruce needle) dominated. In this case, an adaptation of the microbial community to a natural input of low-quality litter could be an explanation (Strickland et al. 2008, Freshet et al. 2012). Besides influencing the microbial community, riparian vegetation may affect the macroinvertebrate assemblage, if shedder species composition is adapted to amounts and types of litter in the standing stock (Murphy and Giller 2000, Leroy and Marks 2006). This is in line with the results showing the fastest shredder-mediated litter decomposition at sites dominated by birch leaf input (Figure 9). Further, results on the influence of litter input during summer and fall suggest that microbes and shredders are affected by the variation in quality of availability litter during the whole year.

4.2 Water chemistry and stream physical characteristics

The variation in litter decomposition in response to water chemistry and stream characteristics, and that the responses differed among litter species, further show the complexity in understanding drivers of the decomposition processes. Nutrient levels have often been shown to increase decomposition (Suberkropp and Cauvert 1995, Suberkropp et al. 2010). My results support this, and that the decomposition processes in boreal headwaters is NO_3 limited (Burrows et al. 2015), but only for alder and birch. The negative influence of the other major nutrient, phosphorus (PO_4), on spruce decomposition can be attributed to PO_4's positive correlation with DOC, which therefore was negatively associated with spruce decomposition. Microbes (primarily bacteria) are the main user of DOC in the water (Hall and Meyer 1998). The observed negative relationship between DOC and spruce decomposition may therefore be explained by microbes using C from the water rather than from the litter. Correspondingly, when poor-quality DOC (high SUVA254) increases, microbes start using leaf litter instead of DOC as energy source, resulting in increased microbial litter decomposition. Hence, food selection in bacteria might change from litter to DOC and vice versa, depending on the quality of leaf litter and DOC. This argument is supported by the fact that DOC only was negatively associated with decomposition of low-quality litter (i.e. spruce), suggesting that DOC is a more preferable energy source than low-quality litter. Overall, these results suggest that the magnitude and effect of increased nutrient input to headwater streams are dependent on type of system, type of litter, quality (and amount) of DOC, and type of nutrient that is increased.

The positive influence of stream velocity on litter decomposition can be due increased litter fragmentation, which not only increases mass loss directly but also indirectly by promoting microbial colonization, and thus decomposition, due to increased surface area (Spänhoff et al. 2007). With higher stream velocity sediment transport increases, further increasing

physical fragmentation of litter (Spänhoff et al. 2007). High toughness in needles could explain why velocity did not affect decomposition of lodgepole pine needles. With the predicted higher precipitation during wet season due to climate change (Whitehead et al. 2009), stream velocity is believed to increase, which may ultimately lead to increased fragmentation and a likely increased litter decomposition.

Lodgepole pine decomposition deviates from the other species, as it was the only species that was influenced by temperature, despite several studies arguing for the importance of temperature for controlling microbial production and, hence, litter decomposition. In an experiment investigating temperature effects on decomposition of litter of different qualities, Fierer et al. (2005) showed that decomposition of low-quality litter was more sensitive to changes in temperature than decomposition of high-quality litter. My results confirm these findings, as lodgepole pine had the lowest litter quality of the four studied species. However, the low variation in mean temperature (0.073 °C) between the streams I studied might not be enough to significantly detect changes in the decomposition of alder, birch and spruce. The effect of increased temperature on litter decomposition is also greatest at low temperatures (Fierer et al. 2005), which is why effects of a small change in temperature may have larger effect depending on what range of temperatures that are investigated. Therefore, in context of climate change, it is important to not only consider a potential increase in mean temperature, but also the temperature range. Nevertheless, my results suggest that increased temperature will increase litter decomposition rates for lodgepole pine litter.

4.3 Adaptation to non-native species

Exotic species often differ in timing, quality and quantity of litter input to aquatic systems compared to native species (Casas et al. 2013). My results show that aquatic systems' ability to decompose non-native species differ compared to native species, even compared to species of similar litter quality (i.e. spruce). Additionally, lodgepole pine had the lowest decomposition rate, which implies a low secondary production and production of FPOM in stream where lodgepole pine needles dominates litter input (i.e. source availability to stream detritivores). Whether the observed differences in litter decomposition of lodgepole pine when compared to native species is due to the fact that it is alien to the streams (and their organisms) or just because the litter is of low quality is hard to tell, because exotic species often produce litter of low quality (Casas et al. 2013). However, results from Canhoto and Graça (1996) showed no difference in decomposition when comparing native and non-native species, but found a significant effect of litter quality, suggesting that litter quality is the main driver. While my results do not refute these findings, the fact that lodgepole pine decomposition was differently associated with the main influencing variables than decomposition of the other low-quality species (i.e. spruce) suggests that other variables than just quality can be important. The continuous use of lodgepole pine in the Swedish forestry, sector may have negative effects on the productivity of headwater ecosystems but also on downstream freshwater and near-shore ecosystems, such as lakes and wetlands

4.4 Variable associations

The positive association between birch input and several variables that in turn were positively related to decomposition of alder, birch, and spruce decomposition shows that riparian vegetation, stream characteristics and water chemistry change simultaneously as the systems undergoes succession. Supporting this is the fact that, after disturbance in the catchment, birch is common in an early-successional forest (Essen et al. 1997), as are increased water velocity, decreased canopy cover, and high discharge of DOC and nutrients (Swank et al. 2001). The positive relationship of SUVA254 to birch litter input, and negative to spruce litter input, indicates that DOC quality may also follow forest succession, with poor-quality DOC in an early-successional forests and high-quality DOC in late succession forests. This is supported by the fact that, with exception of wetlands, the largest proportion of DOC in streams originates from the terrestrial environment from either soil or leached from leaf litter. During fall, after defoliation of leaves, the largest part of DOC originates from leaf litter

(Hongve 1999). Depending on tree species, both amount and content of leached DOC will differ, which affects the water chemistry in the aquatic system (Hongve 1999). This change in amount and content of DOC due to changes in tree species composition further shows how the succession in the terrestrial and aquatic environment is associated.

These relationships between riparian vegetation, water chemistry, and stream characteristics give valuable information for optimization of management methods for riparian zones, to prevent adverse forestry effects on small streams. If the aim is to create buffer zones that minimize negative effects and mimic natural systems after disturbance, riparian vegetation should consist of early-successional tree species. This will create systems that represent water chemistry and riparian vegetation that are needed in order to maintain landscape-scale heterogeneity and habitats with rapid in-stream litter decomposition, high production of FPOM, and high secondary production, which current management often fails to do (Kuglerova et al. 2014).

4.5 Decomposer community and litter decomposition

The higher decomposition rate of deciduous species, compared to coniferous species, was expected, based on result from previous studies (Petersen and Cummins 1974). Since no previous study has compared the exact same species composition and used the same study design in a boreal system, the specific decomposition rates (i.e. *k*) are hard to compare between studies. Nevertheless, the negative relation between litter C/N ratio and decomposition rates confirms the influence by litter stoichiometry. In addition, secondary compounds in litter, which were not measured in this study, could be important.

Levels of ergosterol in birch and alder leaves were similar to those reported by Haapala et al. (2001), but more surprisingly, fungal biomass did not correlate with either litter C/N ratio or decomposition rates. Since fungi are the main contributor to stream microbial composition of litter (Gulis and Suberkrop 2003), it is unlikely that a difference in bacterial biomass can explain why ergosterol levels and decomposition rates are not related. Leaching of soluble compounds from the litter during the first days in the decomposition process has large effects on subsequent biomass loss and quality of remaining litter biomass (Petersen and Cummins 1974). Further, the leaching rate can vary significantly between species (McDowell and Fisher 1976). Therefore, a higher mass loss due to leaching in alder, and subsequent rapid decrease in quality of remaining litter biomass, could explain why alder, despite its low ergosterol content, had decomposed faster than birch. Further, since fungal colonization patterns differ between species (Gessner and Chauvet 1994), the peak in fungal mass occurs at different times, with litter of low quality (i.e. lodgepole pine and spruce) reaching its peak in fungal biomass later. With only one measurement for the whole study period, the risk of underestimating ergosterol levels for species that peak in ergosterol earlier (i.e. primarily alder) is pronounced, and may explain the lack of relationship between ergosterol and decomposition rates in this study. The temporal difference in peak biomass could also explain why only decomposition of lodgepole pine was highly influenced by fungal activity, that presumably show a late peak in ergosterol levels due to its low-quality litter. This is despite the fact that previous studies have shown the importance of fungal activity in litter decomposition.

The generally much higher decomposition rate in coarse- than in fine-mesh bags demonstrate the importance of shredder fragmentation in the decomposition process, as would be expected at high latitudes and hence low water temperatures (e.g. Iron et al. 1994). Previous studies have shown that variation in abundance of shredders is the main explanation for variation in litter decomposition rates (Graca 2001). However, the result from this study showed that shredder species richness and biomass is more important than abundance in decomposition of birch leaves, but less important than some other environmental factors, such as litter input and water chemistry (e.g. Jonsson et al. 2001). The influence of shredders species richness on litter decomposition shows the importance of biodiversity for maintaining the function of ecosystem processes, even if specific shredder species might be more important than diversity *per se* (Robinson et al. 1998). Hence, even if

environmental variables such as litter input and water chemistry are the most important for litter decomposition, loss of species will potentially affect the whole stream network, since it decreases litter decomposition, with consequences for both nutrient cycling and production of FPOM, and subsequent water quality for downstream systems. Maintaining a high biodiversity of shredders should therefore be of great interest to maintain good health and function of headwaters.

Although shredder biomass and species richness were influential, birch litter decomposition was most influenced by variables affecting microbial decomposition. Increased microbial colonization can change litter stoichiometry and make litter more palatable for shredders and their feeding rate more efficient (e.g. McKie and Malmqvist 2009, Fuller et al. 2014), which would explain why variables affecting microbial decomposition is more influential than shredder biomass and species richness when predicting decomposition.

4.6 Litter standing stock

The strong influence by litter input for predicting litter-standing stock showed that litter input is the most important factor for litter standing stock in the streams (Figure 10). Although my calculations predicted that biomass was still left after the study period, a loss of 72% during fall indicates that food shortage may occur during winter or early spring. As this study shows, decomposition of high-quality litter is faster and will disappear first from the litter standing stock. The availability of low-quality litter might therefore be important to prevent food shortage for in-stream biota. This highlights the importance of both low- and high-quality litter for maintaining the productivity of headwater streams during the whole year (e.g. Cummins et al. 1989, Haapala et al. 2001). Naturally, this is achieved by a diverse tree species composition in the riparian zone, created by small-scale disturbance (i.e dying trees), which enhance colonization of early successional species. In the managed forest that typically consist of even age monocultures (Essen et al 1997), a more variable buffer zone can be created by cutting down individual trees to mimic natural disturbance. My estimations only included three of the six litter input categories. Therefore, the standing stock biomass is likely greater, but whether an inclusion of the other categories would influence the rate of loss is harder to predict, since it depends on the litter quality. Still, birch litter input was by far the most important determinant for litter standing stock in the streams.

4.7 Conclusion

The results from this study show that environmental influences on litter decomposition differ among species. This suggests that caution is needed when making general conclusions about how processes might respond to environmental change. Further, the results suggests that laboratory experiments on litter decomposition, where single influencing parameters often are investigated, might not be relevant to natural conditions in which sets of interacting parameters might influence the processes in the same or opposite direction. The observed differences in decomposition rate between species suggests that the species contributing to litter input can have great impact on an aquatic system's secondary production, results that should be considered in management of riparian zones. This study also shows the importance of both microbial and shredder activity in the litter decomposition process. A reduced activity of either can have huge impacts as it changes the energy base for the food web and modifies the transport of particles and nutrients to downstream regions.

Stream food webs seem to be able to adjust to changes in the quality of litter input that naturally occurs as the forest undergoes succession. Supporting this is the fact that streams with riparian vegetation input dominated by early-successional species showed higher decomposition rates of late-successional species litter. However, the non-native species differed from this pattern, once again showing the difficulty to make general predictions for different types of systems and litter. Further, when exotic species are intentionally introduced to terrestrial systems, there is often little concern regarding influences on adjacent aquatic systems. Results from this study show that a non-native tree species can

adversely impact in-stream litter decomposition rates, with likely negative consequences for secondary production and energy transport to downstream aquatic ecosystem.

5 Acknowledgement

Special thanks to my supervisor Micael Jonsson and to Ryan Sponseller for making this project happen. Also a great thanks to Ryan Burrows for field assistance and together with Anna Sundelin and Emma Andersson have given valuable feedback on the report. Further thanks to all the students from the masterstudent-room for help and interesting discussions during the writing process.

6 References

Aerts, R. 1997. Climate, leaf litter chemistry and leaf litter decomposition in terrestrial ecosystems: a triangular relationship. *Oikos* 439-449.

Allan, J.D. 2004. Landscape and riverscapes: the influence of land use on stream ecosystems. *Annual review of ecology, evolution and systematics* 35:257-284.

Bärlocher, F. 1985. The role of fungi in the nutrition of stream invertebrates. *Botanical journal of the Linnean Society* 91:83-94.

Benfield, E. F. 1996. Leaf breakdown in stream ecosystems. In: *Methods in stream ecology,* F. R. Hauer, and G. A. Lambert (eds.), 711-720. Second edition. Burlington: Academic Press.

Bernhardt, E. S., Likens, G. E., Buso, D. C., & Driscoll, C. T. 2003. In-stream uptake dampens effects of major forest disturbance on watershed nitrogen export. *Proceedings of the National Academy of Sciences* 100:10304-10308.

Bishop, K., Buffam, I., Erlandsson, M., Fölster, J., Laudon, H., Seibert, J. and Temnerud, J. 2008. Aqua Incognita: the unknown headwaters. *Hydrological processes* 22:1239-1242.

Burrows, R. M., Hotchkiss, E. R., Jonsson, M., Laudon, H., McKie, B. G. and Sponseller, R. A. 2015. Nitrogen limitation of heterotrophic biofilms in boreal streams. *Freshwater Biology* 60:1237–1251.

Canhoto, C. and Graça, M.A.S. 1996. Decomposition of *Eucalyptus globules* leaves and three native leaf species (*Alnus glutinosa, Castanea sativa* and *Quercus faginea*) in a Portuguese low order stream. *Hydrobiologica* 333:79-85.

Casas, J. J., Larrañaga, A., Menéndez, M., Pozo, J., Basaguren, A., Martínez, A., Pérez, J., González, J. M., Salvador, M., Casado, C., Descals, E., Roblas, N., López-González, J. A. and Valenzuela, J. L. 2013. Leaf litter decomposition of native and introduced tree species of contrasting quality in headwater streams: How does the regional setting matter? *Science of the Total Environment 458*:197-208.

Cummins, K. W., Wilzbach, M. A., Gates, D. M., Perry, J. B., Taliaferro, W. W. 1989. Shredders and riparian vegetation. *BioScience* 39:24-30.

Dangles, O., & Malmqvist, B. 2004. Species richness–decomposition relationships depend on species dominance. *Ecology Letters 7*:395-402.

Engelmark, O., Sjöberg, K., Andersson, B., Rosvall, O., Ågren, G. I., Baker, W. L., Barklund, P., Björkman, C., Despain, D.G., Elfving, B., Ennos, R.A., Karlman, M., Knecht, M.F., Knight, D.G., Ledgard, N.J., Lindelöw, Å., Nilsson, C., Peterken, G.F., Sörlin, S. and Sykes, M. T. 2001. Ecological effects and management aspects of an exotic tree species: the case of lodgepole pine in Sweden. *Forest Ecology and Management* 141:3-13.

Essen, P.A., Ehnström, B., Ericson, L. and Sjöberg, K. 1997. Boreal forests. *Ecological Bulletins* 46:16-47.

Fierer, N., Craine, J. M., McLauchlan, K. and Schimel, J.P. 2005. Litter quality and the temperature sensitivity of decomposition. *Ecology* 86:320-326.

Fisher, S. G., and Likens, G. E. 1973. Energy flow in Bear Brook, New Hampshire: an integrative approach to stream ecosystem metabolism. *Ecological monographs* 43:421-439.

Freshet, G. T., Aerts, R. and Cornelissen, J. H. C. 2012. Multiple mechanisms for trait effects on litter decomposition: moving beyond home-field advantages with a new hypothesis. *Journal of Ecology* 100:619-630.

Fuller, C. L., Evans-White, M. A., and Entrekin, S. A. 2014. Growth and stoichiometry of a common aquatic detritivore respond to changes in resource stoichiometry. *Oecologia* 1-12.

Gessner, M. O., and Chauvet, E. 1994. Importance of stream microfungi in controlling breakdown rates of leaf litter. *Ecology* 75:1807-1817.

Graça, M.A.S. 2001. The Role of Invertebrates on Leaf Litter Decomposition in Streams – a Review. International Review of Hydrobiology 86:383-393.

Gulis, V. and Suberkropp, K. 2003. Leaf litter decomposition and microbial activity in nutrient-enriched and unaltered reaches of headwater stream. *Freshwater Biology* 48:123-134.

Haapala, A., Muotka, T., and Markkola, A. 2001. Breakdown and macroinvertebrate and fungal colonization of alder, birch, and willow leaves in a boreal forest stream. *Journal of the North American Benthological Society* 20:395-407.

Hall, R. O. and Meyer, J. L. 1998. The trophic significance of bacteria in a detritus-based stream food web. *Ecology* 79:1995-2012.

Honve, D. 1999. Production of dissolved organic carbon in forested catchments. *Journal of Hydrology* 224:91-99.

Hulland, J., Ryan, M.J. and Rayner, R.K. 2010. Modeling customer satisfaction: A comparative performance evaluation of covariance strucyure analysis versus partial least squares. In: *Handbook of partial least squares: Concepts, methods and applications*. V. Esposito Vinzi, W. W. Chin, J. Henseler, H. Wang (eds.), 307-325. Berlin-Heidelberg: Springer.

Irons, J. G., Oswood, M. W., Stout, R., & Pringle, C. M. 1994. Latitudinal patterns in leaf litter breakdown: is temperature really important? *Freshwater Biology* 32:401-411.

Jonsson, M., Malmqvist, B., Hoffsten, P.-O. 2001. Leaf litter breakdown rates in boreal streams: does species richness matter? *Freshwater Biology* 46:161-171.

Kempe, G. 2014. Skog och skogsmark, Forest and forestry. In *Skogsstatistisk årsbok, Swedish statistical yearbook of forestry*. L. Christiansesn (eds.), 41-70. Jönköping: Swedish forest agency.

Kominoski, J.S., Marczak, L. B. and Richardson, J.S. 2011. Riparian forest composition affects stream litter decomposition despite similar microbial and invertebrate communities. *Ecology* 92:151-159.

Kuglerová, L., Ågren, A., Jansson, R. and Laudon, J. 2014. Towards optimizing riparian buffer zones: ecological and biogeochemical implications for future management. *Forest ecology and management* 334:74-84.

Leroy, C. J. and Marks, J. C. 2006. Litter quality, stream characteristics and litter diversity influence decomposition rates and macroinvertebrates. *Freshwater Biology* 51:605-617.

Manning, D. W. P., Rosemond, A. D., Kominoski, J. S., Gulis, V., Benstead, J. P., and Maerz, J. C. 2015. Detrital stoichiometry as a critical nexus for the effects of streamwater nutrients on leaf litter breakdown rates. *Ecology*. http://dx.doi.org/10.1890/14-1582.1

McDowell, W. H. and Fisher, S. G. 1976. Autumnal processing of dissolved organic matter in a small woodland stream ecosystem. *Ecology* 57:561-569.

McKie, B. G. and Malmqvist, B. 2009. Assessing ecosystem functioning in streams affected by forest management: increased leaf decomposition occurs without changes to the composition of benthic assemblage. *Freshwater Biology* 54:2086-2100.

Meyer, J.L., Edwards, R.T. and Risley, R. 1987. Bacterial growth on dissolved organic carbon from a blackwater river. *Microbial ecology* 13:13-29.

Meyer, J. L., Strayer, D. L., Wallace, J. B., Eggert, S. L., Helfman, G. S., & Leonard, N. E. 2007. The contribution of headwater streams to biodiversity in river networks. *Journal of the American water resource association* 43:86-103.

Murphy, J. F. and Giller, P. S. 2000. Seasonal dynamics of macroinvertebrate assemblage in the benthos and associated with detritus packs in two low-order streams with different riparian vegetation. *Freshwater biology* 43:617-631.

Perdue, E. M. 1998. Chemical composition, structure, and metal binding properties. In *Aquatic humic substances,* D. O. Hessen and L. J. Tranvik (eds.), 41-61. Berlin Heidelberg: Springer.

Petersen, P.C. and Cummins, K.W. 1974. Leaf processing in a woodland stream. *Freshwater Biology* 4:343-368.

R Core Team. 2012. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

Robinson, C.T., Gessner, M. O. and Ward, J. V. 1998. Leaf breakdown and associated macroinvertebrates in alpine glacial streams. *Freshwater Biology* 40:215-228.

Rosemond, A. D., C. M. Swan, J. S. Kominoski, and S. E. Dye. 2010. Non-additive effects of litter mixing are suppressed in a nutrient-enriched stream. *Oikos* 119:326-336.

Schmidt-Kloiber A. and Hering D. (eds.) 2012. www.freshwaterecology.info - the taxa and autecology database for freshwater organisms, version 5.0. 2014-12-10.

Spänhoff, B., Augspurger, C., Keusel, K., 2007. Comparing field and laboratory breakdown rates of coarse particulate organic matter: sediment dynamics mask the impacts of dissolved nutrients on CPOM mass loss in streams. *Aquatic Science* 69:495-502.

Stenroth, K., Polvi, L.E., Fältström, E. and Jonsson, M. 2015. Land-use effects on terrestrial consumers through changed size structure of aquatic insects. *Freshwater Biology* 60:136-149.

Strickland, M. S., Osburn, E., Lauber, C., Fierer, N., & Bradford, M. A. 2009. Litter quality is in the eye of the beholder: initial decomposition rates as a function of inoculum characteristics. *Functional Ecology 23*:627-636.

Swank, W. T., Vose, J. M. and Elliot, K.J. 2001. Long-term hydrologic and water quality responses following commercial clearcutting of mixed hardwoods on a southern Appalachian catchment. *Forest Ecology and Management* 143:163-178.

Suberkropp, K. and Cauvert, E. 1995. Regulation of leaf breakdown by fungi in streams: influence of water chemistry. *Ecology* 76:1433-1445.

Suberkropp, K., Gulis, V., Rosemond, A. D., and Benstead, J. P. 2010. Ecosystem and physiological scales of microbial responses to nutrients in a detritus-based stream: Results of a 5 year continuous enrichment. *Limnology and Oceanography* 55:149-160.

Temnerud J, Bishop K. 2005. Spatial variation of stream water chemistry in two Swedish boreal catchments: implications for environmental assessment. *Environmental Science and Technology* 39:1463-1469.

Wallace, J.B., Eggert, S. L., Meyer, J. L. and Webster, J. R. 1999. Effects of resource limitation on a detrital-based ecosystem. *Ecologiacal monographs* 69:409-422.

Wallace, T. A., Ganf, G. G. and Brookes, J. D. 2008. A comparison of phosphorus and DOC leachates from different types of leaf litter in an urban environment. *Freshwater Biology* 53:1902-1913.

Webster, J.R and Benfield, E.F. 1986. Vascular plant breakdown in freshwater ecosystems. *Annual review of ecology and systematics* 17:567-94.

Webster, J. R., Golladay, S. W., Benfield, E. F., Meyer, J. L., Swank, W. T., and Wallace, J. B. 1992. Catchment disturbance and stream response: an overview of stream research at Coweeta Hydrologic Laboratory. *River conservation and management* 15:232-253.

Weishaar, J. L., Aiken, G. R., Bergamaschi, B. A., Fram, M. S., Fujii, R., and Mopper, K. 2003. Evaluation of specific ultraviolet absorbance as an indicator of the chemical composition and reactivity of dissolved organic carbon. *Environmental Science and Technology* 37:4702-4708.

Wipfli, M.S. and Gregovich, D.P. 2002. Export of invertebrates and detritus from fish free headwater streams in southeastern Alaska: implication for downstream salmonid production. *Freshwater Biology* 47:957-969.

Whitehead, P. G., Wilby, R. L., Battarbee, R. W., Kernan, M., & Wade, A. J. 2009. A review of the potential impacts of climate change on surface water quality. *Hydrological Sciences Journal* 54:101-123.

Appendix 1

Table 1. Mean values for the water chemistry parameters during the study period. Values in parentheses are variance between the measurements.

Site	PO_4 (µg/L)	NH_4 (µg/L)	NO_3 (µg/L)	DIN (µg/L)	TN (mg/L)	DOC (mg/L)	SUVA254 ($L/m^{-1}mg^{-2}$)	Conductivit (µS/cm^3)	pH
B1	3.20 (0.72)	27.47 (8.35)	12.10 (1.57)	39.57 (9.88)	0.33 (0.05)	19.71 (4.52)	5.42 (1.76)	27.48 (1.63)	5.57 (0.35)
BCC	6.03 (0.63)	12.79 (1.47)	60.04 (12.41)	72.82 (12.89)	0.41 (0.05)	18.93 (3.45)	5.50 (1.82)	20.93 (2.63)	5.13 (0.30)
BF	5.04 (0.52)	7.12 (0.73)	7.00 (0.25)	14.12 (0.95)	0.37 (0.05)	19.55 (2.40)	4.96 (1.14)	22.62 (2.23)	5.43 (0.21)
G1	3.11 (0.77)	5.66 (0.76)	5.73 (0.46)	11.39 (1.14)	0.28 (0.05)	17.07 (3.39)	6.39 (2.35)	15.67 (5.39)	5.32 (0.31)
G2	1.06 (0.26)	7.33 (1.15)	8.07 (1.26)	15.40 (2.36)	0.27 (0.01)	12.97 (0.89)	4.14 (0.45)	22.32 (7.29)	6.40 (0.06)
G3	1.26 (0.37)	3.65 (0.30)	7.00 (0.66)	10.64 (0.36)	0.15 (0.02)	6.85 (1.03)	5.47 (1.58)	15.17 (5.14)	6.46 (0.09)
K1	6.12 (0.71)	5.46 (0.93)	7.52 (1.94)	12.99 (2.51)	0.16 (0.04)	9.23 (3.31)	8.26 (5.37)	19.93 (3.32)	6.24 (0.30)
K5	1.43 (0.36)	4.75 (0.69)	11.48 (4.05)	16.23 (3.73)	0.17 (0.01)	7.61 (0.54)	4.26 (0.68)	17.73 (6.02)	6.10 (0.12)
K8	2.51 (0.14)	11.22 (1.44)	17.69 (1.89)	28.91 (2.68)	0.35 (0.04)	19.24 (2.99)	5.86 (1.91)	15.84 (5.38)	6.00 (0.24)
KL	2.40 (0.23)	8.13 (1.83)	8.41 (1.25)	16.54 (3.02)	0.41 (0.10)	23.56 (6.38)	6.11 (2.35)	24.24 (2.41)	5.35 (0.36)
Kr1	2.64 (0.39)	20.66 (3.16)	28.14 (2.77)	48.79 (4.39)	0.43 (0.07)	23.15 (4.45)	5.27 (1.55)	29.47 (1.12)	5.45 (0.23)
Kr6	3.53 (0.48)	16.46 (2.45)	16.81 (2.64)	33.27 (5.05)	0.35 (0.02)	18.13 (1.51)	5.24 (0.76)	19.37 (1.02)	5.57 (0.24)
Kr7	9.87 (2.23)	11.87 (1.08)	16.49 (1.43)	28.36 (1.36)	0.44 (0.07)	27.32 (5.17)	5.24 (1.48)	30.21 (2.35)	4.92 (0.29)
Röd	2.00 (0.34)	9.33 (2.04)	18.80 (5.02)	28.12 (7.01)	0.37 (0.08)	17.74 (3.78)	4.97 (1.57)	29.27 (1.28)	6.23 (0.23)
S2	11.73 (1.06)	11.26 (1.01)	9.20 (0.36)	20.46 (1.32)	0.47 (0.02)	35.73 (2.82)	4.24 (0.42)	39.07 (3.41)	4.95 (0.37)
S26	1.53 (0.09)	4.30 (0.52)	8.80 (1.83)	13.09 (2.08)	0.23 (0.04)	10.27 (1.57)	4.44 (1.23)	38.96 (1.37)	6.65 (0.12)
S3	3.95 (0.78)	10.91 (1.17)	7.88 (0.68)	18.79 (0.87)	0.41 (0.04)	21.26 (2.24)	4.69 (0.93)	28.53 (3.43)	5.42 (0.26)
S6	1.90 (0.23)	5.92 (0.19)	16.75 (4.58)	22.67 (4.75)	0.34 (0.01)	15.24 (0.49)	3.95 (0.22)	44.28 (4.11)	5.70 (0.10)
V1	2.10 (0.43)	7.16 (1.77)	14.25 (3.34)	21.42 (4.78)	0.32 (0.11)	17.22 (6.48)	6.90 (3.07)	28.33 (1.42)	6.20 (0.27)
V2	1.76 (0.16)	5.56 (1.04)	9.20 (1.78)	14.76 (2.38)	0.28 (0.09)	15.12 (5.15)	6.32 (2.62)	26.99 (1.15)	6.17 (0.26)

Appendix 2

Table 1. Shredder species composition for the sites. Values are mean values for each litter-bag.

	Site																			
Species	B1	Bcc	BF	G1	G2	G3	K1	K5	K8	KL	Kr1	Kr6	Kr7	Rod	S2	S26	S3	S6	V1	V2
PLECOPTERA																				
Nemurella picteti	7	38	2	8	-	9	32	4	3	8	4	6	4	1	23	1	-	-	-	-
Nemoura avicularis	-	-	1	-	2	-	-	-	1	-	-	-	-	1-	-	-	-	-	5	15
Nemoura cinerea	306	522	64	108	37	6	339	5-	138	220	112	306	128	33	693	13	284	225	6	9
Nemoura flexuosa	1	-	-	-	15	-	-	-	-	11	-	-	-	79	-	-	-	-	32	27
Protonemura meyeri	-	-	-	-	23	-	-	-	-	1	-	-	-	6	-	-	-	-	6	-
Amphinemura sulcicollis	-	-	-	-	12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Taeniopteryx nebulosa	-	-	3	1	2	-	-	-	-	-	-	-	-	6	-	-	-	-	8	8
Leuctra digitata	-	-	14	-	2	-	3	1	-	21	-	-	-	19	-	-	-	-	17	11
Leuctra nigra	3	-	-	-	-	1	6	2	5	1	10	1	1	-	2	1	-	-	-	-
Capnopsis schilleri	-	-	-	-	-	-	-	-	-	-	-	-	-	11	-	-	-	-	-	-
TRICHOPTERA																				
Chaetopteryx villosa	-	2	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-
Potamophylax nigricornis	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5	-	-	-	-
Potamophylax cingulatus	-	-	2	-	-	1	-	-	-	2	-	-	-	2	-	-	-	-	-	-
Potamophylax cingulatus latipennis	-	1	23	-	-	8	6	-	1	23	-	2	-	-	-	-	-	-	3	7
Micropterna lateralis	-	1	-	-	-	-	-	7	-	-	-	-	-	-	2	-	-	-	-	-
Micropterna sequax	-	1	-	-	-	-	-	-	-	-	-	-	-	-	2	24	-	5	-	-
Micropterna sp.	-	-	-	-	-	-	-	5	-	1	5	-	1	-	-	-	-	-	-	-
Limnephilidae	-	-	2	2	-	-	-	-	1	1	-	1	-	-	-	-	-	-	-	1
COLEOPTERA																				
Helophorus	-	1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Appendix 3

Table 1. Mean values of mass input (g) of different litter categories to litter trays in summer and fall

	Summer							Fall						
Site	Alder	Birch	Other	Pine	Spruce	Twigs	Tot	Alder	Birch	Other	Pine	Spruce	Twig	Tot
B1	0.22	0.98	1.58	0.12	4.04	0.32	7.26	1.57	9.55	1.01	0.27	2.78	5.83	21.01
BCC	0.00	0.19	1.01	0.03	0.03	0.06	1.32	0.00	6.42	1.30	0.50	0.16	0.00	8.37
BF	0.00	0.65	1.03	0.05	1.15	0.76	3.64	0.00	8.73	2.39	0.48	3.60	0.44	15.64
G1	0.00	0.62	1.04	0.50	0.03	0.76	2.96	0.00	23.34	4.70	0.18	0.21	0.63	29.06
G2	0.00	0.28	1.51	0.94	1.22	0.49	4.44	0.57	4.50	1.93	1.68	5.62	0.32	14.63
G3	0.14	0.22	0.78	0.01	1.62	1.30	4.08	1.57	4.66	1.20	0.05	4.37	0.25	12.10
K1	0.08	0.89	1.42	0.30	0.10	0.71	3.49	0.04	35.72	1.72	0.90	0.39	0.49	39.26
K5	0.12	2.70	1.13	0.17	0.08	0.01	4.21	3.55	9.78	1.61	1.80	0.63	0.64	18.00
K8	0.01	1.66	2.38	0.00	2.18	1.96	8.19	1.87	28.99	3.93	0.00	5.78	0.39	40.95
KL	1.10	1.67	0.73	0.00	0.01	0.57	4.08	7.90	6.91	2.06	0.00	4.83	0.31	22.01
Kr1	0.00	0.60	1.19	0.23	1.75	1.38	5.16	0.00	8.89	0.54	1.34	2.94	0.19	13.89
Kr6	0.00	0.00	1.24	0.04	2.61	1.24	5.13	0.00	1.14	1.42	0.11	6.60	0.45	9.71
Kr7	0.01	0.72	1.26	0.01	2.82	0.70	5.52	0.00	9.69	0.35	0.04	8.82	0.19	19.09
Rod	0.61	0.16	0.81	0.00	1.68	0.17	3.43	1.94	32.12	6.20	0.04	0.02	0.50	40.82
S2	0.00	0.63	2.74	3.45	4.08	0.87	11.77	0.33	12.72	1.58	7.74	9.53	1.42	33.32
S26	0.00	0.22	1.90	0.47	2.05	0.41	5.07	0.00	7.76	2.35	2.46	8.97	0.66	22.19
S3	0.02	1.29	1.04	1.76	0.45	1.05	5.62	0.89	24.79	1.23	2.44	2.19	0.58	32.12
S6	1.76	0.29	2.68	0.00	0.36	3.26	8.35	23.01	13.67	5.49	0.06	1.43	1.15	44.80
V1	0.00	1.10	0.94	0.01	0.70	1.31	4.06	0.61	11.28	3.11	0.13	2.87	0.08	18.09
V2	0.00	1.74	1.05	0.01	0.12	0.39	3.31	0.45	23.64	5.71	0.06	0.41	0.42	30.69

Dept. of Ecology and Environmental Science (EMG)
S-901 87 Umeå, Sweden
Telephone +46 90 786 50 00
Text telephone +46 90 786 59 00
www.umu.se